

DETECTING CRISES, JUMPS, AND CHANGES IN REGIME

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Abstract: Automated model selection and impulse indicator saturation contribute two useful tools to a coherent empirical framework for detecting crises, jumps, and changes in regime. This paper proposes extensions to impulse indicator saturation that may provide more power to detect breaks in common economic situations. Empirical applications include seasonally unadjusted UK narrow money demand, U.S. equity prices, U.K. exchange rates, the U.S. Taylor rule, and forecasts of government debt.

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1 Introduction

The recent financial crisis and Great Recession have highlighted the pervasiveness of structural breaks in the economy. Structural breaks may be rapid, as with crises (financial and otherwise), jumps (as with asset-price volatility), and changes in regime (Arab spring, ...). They may also be more gradually evolving, as with some forms of innovation and globalization. Detection of breaks is important; and recent events emphasize the need for reliable “early-warning” and “early-detection” methods for assessing the state of the economy. Whether detected or not, breaks pose substantive challenges to economic and statistical inference, forecasting, and policy. When do breaks occur, where in the economy, how often, for how long, and of what magnitude?

Automated model selection and impulse indicator saturation contribute two useful tools in a coherent empirical framework for detecting structural breaks such as crises, jumps, and changes in regime. The current paper proposes *extensions* to impulse indicator saturation that may provide more power to detect breaks in common economic situations. Empirical applications include seasonally unadjusted UK narrow money demand, U.S. equity prices, U.K. exchange rates, the U.S. Taylor rule, and forecasts of government debt. Both automated model selection and impulse indicator saturation can demonstrate the robustness of a model to a wide range of feasible alternatives. Those tools also can yield statistical and economic improvements to that model and, in so doing, provide insights into the practical justification of empirical evidence in macro-economics.

This paper is organized as follows. Section 2 summarizes the computer-automated model selection algorithm in Autometrics. That algorithm is central to implementing impulse indicator saturation. Section 3 discusses impulse indicator saturation (IIS) and proposes several extensions that are potentially useful for detecting crises, jumps, and changes in regime. The next three sections illustrate IIS-type techniques. Using Autometrics, Section 4 evaluates the potential search-path dependence of the empirical model of money demand in Ericsson, Hendry, and Tran (1994). Section 4 also employs impulse indicator saturation and some of its extensions to test for parameter constancy in that money demand model; and it develops a new empirical model of UK narrow money demand using the methods in Sections 2 and 3. Section 5 extends IIS results in Ericsson (2011a) and Ericsson and Reisman (2012a) for the global vector autoregressive model in Déés, di Mauro, Pesaran, and Smith (2007), focusing on equations for the U.S. equity market and the U.K. exchange rate. Section 6 summarizes two additional recent applications: the U.S. Taylor rule, and forecasts of the Federal debt. Section 7 offers a few general remarks on empirical evidence in macro-econometric modeling.

2 Computer-automated Model Selection

In economics, data mining has traditionally been viewed as a pejorative activity—a necessary evil, with nominal critical values affected; see Lovell (1983), Denton (1985), and the extensive literature on pre-testing. By contrast, data mining can be constructive rather than pejorative; see Campos and Ericsson (1999). This section thus summarizes an approach to constructive data mining as implemented in the model selection algorithm Autometrics. Section 3 discusses indicator saturation techniques, which utilize that algorithm in Autometrics. Together, that algorithm and indicator saturation techniques can aid in detecting crises, jumps, and changes in regime, as Sections 4 and 6 show.

Hoover and Perez (1999) proposed an automated general-to-specific model-selection algorithm that incorporated many of the features of the “Hendry” or LSE methodology. Hendry and Krolzig (2001) developed a second-generation algorithm called PcGets, which extended and improved upon Hoover and Perez’s algorithm; see also Hendry and Krolzig (1999, 2003, 2005) and Krolzig and Hendry (2001). Doornik and Hendry (2009) implement a third-generation algorithm called Autometrics, which is part of PcGive version 13. Autometrics utilizes one-step and multi-step simplifications along multiple paths following a tree search method. Diagnostic tests serve as additional checks on the simplified models, and encompassing tests resolve multiple terminal models. Both analytical and Monte Carlo evidence show that the resulting model selection is relatively non-distortionary for Type I errors. At an intuitive level, Autometrics functions as a series of sieves that aim to retain parsimonious congruent models while discarding both noncongruent models and over-parameterized congruent models. This feature of the algorithm is eminently sensible, noting that the data generation process itself is congruent and is as parsimonious as feasible. At a more formal level, White (1990, 2009) provides a statistical theoretic basis for this model-building strategy in which diagnostic test statistics (including encompassing statistics) are treated as explicit design criteria. See also Hansen, Lunde, and Nason (2011) for a related discussion on model confidence sets.

The current section summarizes Autometrics as an automated model-selection algorithm, thereby providing the necessary background for interpreting subsequent empirical applications. For ease of reference, the algorithm is divided into three “stages”, denoted Stage 0, Stage 1, and Stage 2. For full details of Autometrics’s algorithm, see Doornik (2008, 2009) and Doornik and Hendry (2009). Hendry and Krolzig (2003) and Campos, Ericsson, and Hendry (2005) describe the relationship of the general-to-specific approach to other modeling approaches in the literature, and Hoover and Perez (2004) extend the general-to-specific approach to cross-section regressions.

Stage 0: the general model, impulse indicator saturation, and F pre-search tests. Stage 0 involves three parts: the estimation and evaluation of the general model,

inclusion of impulse indicator dummies for all observations, and some pre-search tests aimed at simplifying the general model before instigating formal multi-path searches.

First, the general model is estimated, and diagnostic statistics are calculated for it. If any of those diagnostic statistics is unsatisfactory, the modeler must decide what to do next—whether to “go back to the drawing board” and develop another general model, or to continue with the simplification procedure, perhaps ignoring the offending diagnostic statistic or statistics.

Second, and optionally, Autometrics performs block additions and searches of impulse indicator dummies for all observations in a process known as impulse indicator saturation. Doing so generates a robust regression estimator, and it provides a check for parameter constancy. Section 3.1 discusses impulse indicator saturation further.

Third, and also optionally, Autometrics attempts to drop various sets of potentially insignificant variables. Autometrics does so by dropping all variables at a given lag, starting with the longest lag. Autometrics also does so by ordering the variables by the magnitude of their t -ratios and either dropping a group of individually insignificant variables or (alternatively) retaining only a group of individually statistically significant variables. In effect, an F pre-search test for a group of variables is a single test for multiple simplification paths, a characteristic that helps control the costs of search. If these tests result in a statistically satisfactory reduction of the general model, then that new model is the starting point for Stage 1. Otherwise, the general model itself is the starting point for Stage 1.

Stage 1: a multi-path encompassing search. Stage 1 tries to simplify the model from Stage 0 by searching along multiple paths, ensuring that the diagnostic tests are not rejected. If all variables are individually statistically significant, then the initial model in Stage 1 is the final model. If some variables are statistically insignificant, then Autometrics tries deleting those variables to obtain a simpler model. If a simplification is rejected, Autometrics backtracks along that simplification path to the most recent previous acceptable model and then tries a different simplification path. A terminal model results if the model’s diagnostic statistics are satisfactory and if no remaining regressors can be deleted.

If Autometrics obtains only one terminal model, then that model is the final model. However, because Autometrics pursues multiple simplification paths in Stage 1, Autometrics may obtain multiple terminal models. To resolve such a situation, Autometrics creates a union model from those terminal models and tests each terminal model against that union model. Autometrics then creates a new union model, which nests all of the surviving terminal models; and that union model is passed on to Stage 2.

Stage 2: another multi-path encompassing search. Stage 2 in effect repeats Stage 1 (possibly iteratively) by applying the simplification procedures from Stage 1 to the union model obtained at the end of Stage 1. The resulting model is the final model. If Stage 2 obtains more than one terminal model after applying encompassing tests,

then the final model is selected by using the Akaike, Schwarz, and Hannan–Quinn information criteria. See Akaike (1973, 1981), Schwarz (1978), and Hannan and Quinn (1979) for the design of these information criteria, and Atkinson (1981) for the relationships between them.

In addition to optional pre-search simplification and impulse indicator saturation, the algorithm requires several other choices: in particular, target size and fixity of regressors.

Target size. Autometrics requires the modeler to choose which tests are calculated and to specify the critical values for those tests. In principle, the modeler can choose the test statistics and their critical values directly, although doing so is tedious because of the number of statistics involved. To simplify matters, Autometrics offers several options for the “target size”, which incorporates pre-designated selections of test statistics and critical values.

The “target size” is meant to equal “the proportion of irrelevant variables that survives the [simplification] process” (Doornik, 2009, p. 100). In the analyses below, Autometrics’s target size is either 5% or 1%—which are values that appear to approximate the liberal and conservative strategies in PcGets—or 0.1% or smaller (very stringent). The liberal strategy errs on the side of keeping some variables, even although they may not actually matter. The conservative strategy keeps only variables that are clearly significant statistically, erring in the direction of excluding some variables, even although those variables may matter. Which strategy is preferable depends in part on the data themselves, in part on the class of regressors examined (and, in particular, whether indicator saturation is considered), and in part on the objectives of the modeling exercise. Also, the two approaches *may* generate similar or identical results, as seen below.

Fixity of regressors. Autometrics offers the option of designating certain variables as “fixed”. Fixed variables are forced to always be included in regression, whereas free variables (variables that are not fixed) may be deleted by the algorithm.

In short, Autometrics is general-to-specific, multi-path, iterative, and encompassing, with diagnostic tests providing additional assessments of statistical adequacy, and with options for pre-search simplification and robustification. Autometrics can be characterized as having two components:

1. Estimation and diagnostic testing of the general unrestricted model (Stage 0);
and
2. Selection of the final model by
 - (a) pre-search simplification of the general unrestricted model (Stage 0),
 - (b) impulse indicator saturation (Stage 0), and
 - (c) multi-path (and possibly iterative) selection of the final model (Stages 1 and 2).

3 Indicator Saturation Techniques

Subsection 3.1 discusses impulse indicator saturation (IIS) as a procedure for model evaluation, and in particular as a procedure for testing parameter constancy. Subsection 3.2 proposes extensions to impulse indicator saturation.

3.1 Impulse Indicator Saturation

Impulse indicator saturation provides a general procedure for analyzing a model's constancy. Specifically, IIS is a generic test for an unknown number of breaks, occurring at unknown times, with unknown duration and magnitude, anywhere in the sample. Hendry (1999) proposes IIS as a procedure for testing parameter constancy; see Hendry, Johansen, and Santos (2008), Doornik (2009), Johansen and Nielsen (2009), Hendry and Santos (2010), and Ericsson and Reisman (2012a, 2012b) for further discussion and recent developments.

Impulse indicator saturation uses zero-one impulse indicator dummies to analyze properties of a model. There are T such dummies, so inclusion of all T dummies in a model is infeasible. However, blocks of dummies can be included, and that insight provides the basis for IIS. This subsection motivates IIS with two simple Monte Carlo examples, which lead to the generic approach in IIS.

Example 1. This example illustrates the behavior of IIS when the model is correctly specified. Suppose that the DGP for variable w_t is:

$$w_t = \mu_0 + \varepsilon_t \quad \varepsilon_t \sim NID(0, \sigma^2), \quad (1)$$

and that the model estimated is a regression of w on an intercept. That is, w_t is normally and independently distributed with a nonzero mean, and the model is correctly specified. Figure 1a plots Monte Carlo data for a DGP with ($\mu_0 = 20$, $\sigma^2 = 1$, $T = 100$); and Figure 1b plots the estimated model's residuals.

Now, re-estimate the model, but including impulse indicator dummies for the first half of the sample, as represented by Figure 2a. (That estimation is equivalent to estimating the model over the second half of the sample, ignoring the first half.) Drop all statistically insignificant impulse indicator dummies and retain the statistically significant dummies (Figure 2b). Repeat this process, but start by including impulse indicator dummies for the *second* half of the sample (Figure 2d) and retain the significant ones (Figure 2e). Now, re-estimate the original model, including all dummies retained in the two block searches (Figure 2g), and select the statistically significant dummies out of that combined set (Figure 2h). Hendry, Johansen, and Santos (2008) and Johansen and Nielsen (2009) have shown that, under the null hypothesis of correct specification, the fraction of impulse indicator dummies retained is roughly αT , where α is the target size. In Figure 2h, four dummies are retained, whereas $\alpha T = 5\% \cdot 100 = 5$, a close match.



Figure 1: Actual and fitted values and the corresponding scaled residuals for the estimated model, when the DGP does not have a break.

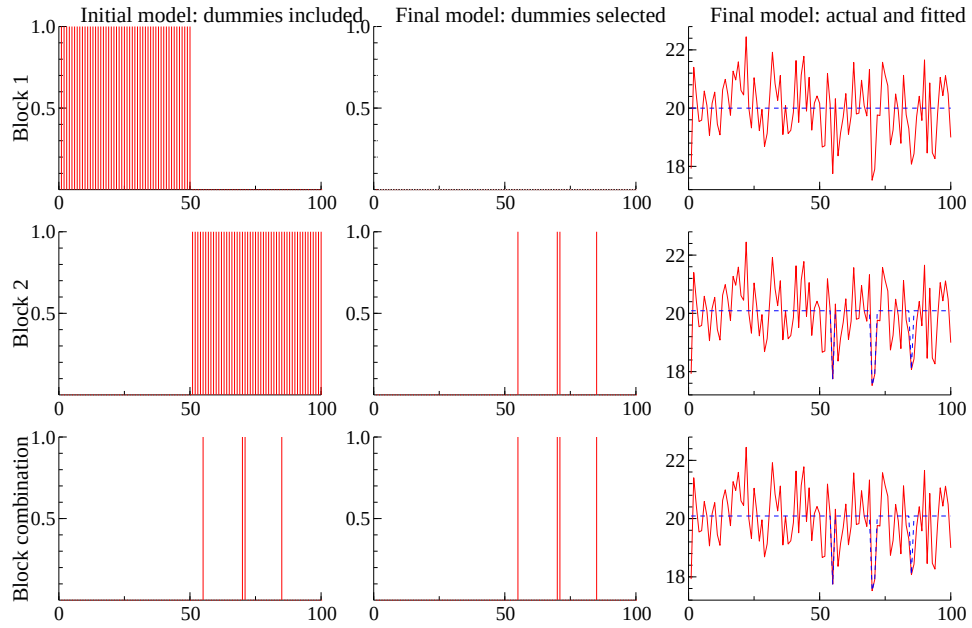


Figure 2: A characterization of split-sample impulse indicator saturation when the DGP does not have a break.

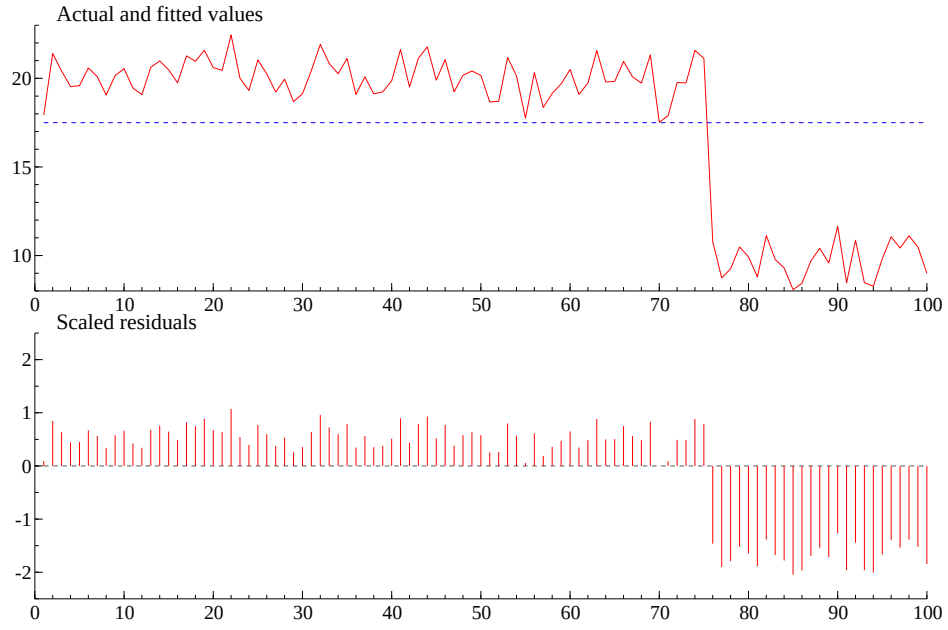


Figure 3: Actual and fitted values and the corresponding scaled residuals for the estimated model, when the DGP has a break.

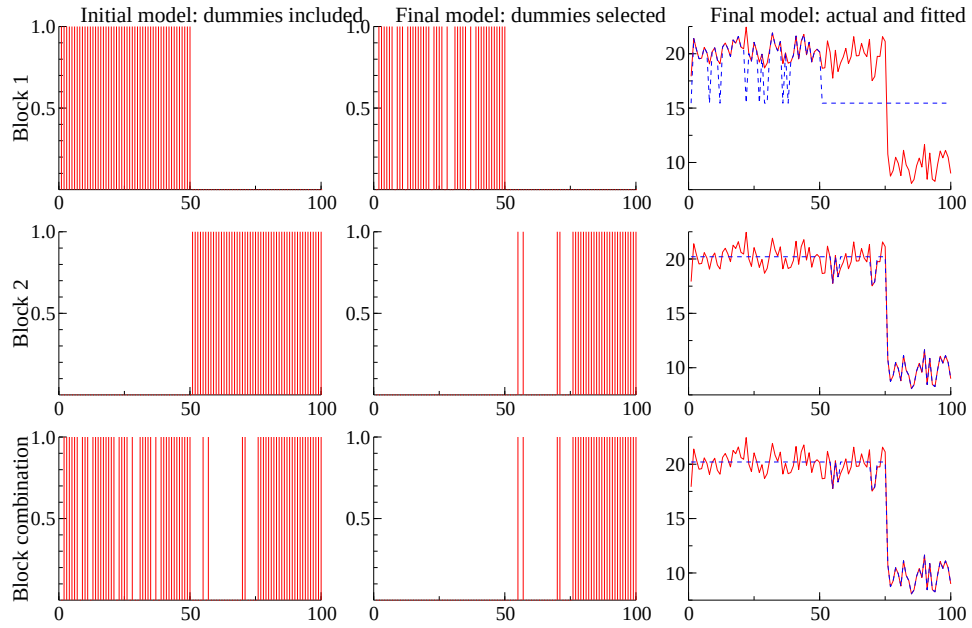


Figure 4: A characterization of split-sample impulse indicator saturation when the DGP has a break.

Example 2. This example illustrates the behavior of IIS when there is an unmodeled break. Suppose that the DGP for variable w_t is:

$$w_t = \mu_0 + \mu_1 S_{76t} + \varepsilon_t \quad \varepsilon_t \sim NID(0, \sigma^2), \quad (2)$$

where $S_{76t} = 0$ ($T = 1, \dots, 75$), $S_{76t} = 1$ ($T = 76, \dots, 100$), and the model estimated is a regression of y on an intercept, ignoring the break S_{76t} . As in Example 1, w_t is normally and independently distributed with a nonzero mean. However, that mean alters over time. The model does not incorporate that change in mean (aka a “location shift”) and so is mis-specified. Figure 3a plots Monte Carlo data for a DGP with ($\mu_0 = 20$, $\mu_1 = -10$, $\sigma^2 = 1$, $T = 100$); and Figure 3b plots the estimated model’s residuals. Interestingly, no residuals lie outside the estimated 95% confidence region, even though the break is 10σ .

Figure 4 plots the corresponding graphs for split-sample impulse indicator saturation. The procedure has high power to detect the break, even although the nature of the break was not utilized in the procedure itself.

In practice, IIS in Autometrics utilizes many blocks, and the partitioning of the sample into blocks may vary over iterations of searches. That said, The specific algorithm for IIS can “make or break” IIS’s usefulness; cf. Doornik (2009) and Castle, Fawcett, and Hendry (2010). IIS is a statistically valid procedure for integrated, cointegrated data; see Johansen and Nielsen (2009). IIS can serve as a diagnostic statistic, and it can aid model development, as discussed in Ericsson (2011a).

Many existing procedures can be interpreted as “special cases” of IIS in that they represent particular algorithmic implementations of IIS. Such special cases include recursive estimation, rolling regression, the Chow (1960) predictive failure statistic (including the 1-step, breakpoint, and forecast versions implemented in OxMetrics), the Andrews (1993) unknown breakpoint test, the Bai and Perron (1998) multiple breakpoint test, tests of nonlinearity, intercept correction (in forecasting), and robust estimation. IIS thus provides a general and generic procedure for analyzing a model’s constancy.

Algorithmically, IIS also solves the problem of having more potential regressors than observations by testing and selecting over blocks of variables. That approach aids interpretation of dynamic factor analysis, principal components, data aggregation, and similar procedures; and it leads to additional tests of parameter constancy. Section 3.2 proposes some extensions to IIS, and it re-interprets some existing tests as IIS-type tests, and it shows how IIS-type tests can be utilized to test for completely general time-varying forecast bias. Sections 4 and 6 consider several applications of IIS.

3.2 Extensions

IIS is a powerful empirical tool for evaluating and improving existing empirical models. This subsection proposes several extensions to IIS that may be more suitable than IIS itself for detecting crises, jumps, and changes in regime. “Extended IIS” also provides a conceptual framework for interpreting existing tests of parameter constancy.

Table 1 summarizes IIS, some other existing tests, and some extensions of IIS, all in terms of the variables involved. Throughout, T is the sample size, t is the index for time, i is the index for indicators, k is the index for economic variables (which are denoted x_{kt}), and K is the total number of potential regressors considered.

A few remarks may be helpful for interpreting the entries in Table 1.

- 1a. **Impulse indicator saturation.** This is the standard IIS procedure proposed by Hendry (1999), with selection among the T zero-one impulse indicators $\{I_{it}\}$.
- 1b. **Super saturation.** In addition to searching across $\{I_{it}\}$, super saturation searches across all possible one-off step functions $\{S_{it}\}$. Step functions are of economic interest because they may capture permanent or long-lasting changes in regime that are otherwise not incorporated into an empirical model. Statistically and numerically, a step function is a parsimonious representation of a sequential subset of impulse indicators that have equal coefficients. Hendry and Pretis (2012) investigate the statistical properties of a closely related saturation estimator—step indicator saturation (SIS), i.e., for only the variables $\{S_{it}\}$.
- 1c. **Ultra saturation.** Partial sums of the partial sums of impulse indicators may also be of economic interest, as those double-sums are broken linear trends $\{T_{it}\}$. Ultra saturation (earlier, sometimes called “super-duper” saturation) searches across $\{I_{it}, S_{it}, T_{it}\}$. Obvious extensions are broken quadratic trends, broken cubic trends, and so forth.
- 1d. **Sequential pairwise impulse indicator saturation.** Extensions 1b and 1c are based on partial sums of the impulse indicators and step functions over the remaining sample, i.e., for all $i \geq t$. Partial sums also can be constructed over fixed-length windows of impulse indicators. The simplest case is sequential pairwise IIS, in which sequential pairs of impulse indicators are added together, i.e., $P_{it} = I_{it} + I_{i+1,t}$. Pairs (or triplets, or quadruplets, etc.) may parsimoniously capture effects that are persistent but not permanent. Nonsequential groupwise IIS is also an option, with some nonsequential groups being of particular interest, such as groups at a seasonal frequency.
- 1e. **Zero-sum pairwise IIS.** Differences of impulse indicators may capture “zero-sum” effects, with $Z_{it} = \Delta I_{it}$, and leading to zero-sum pairwise IIS. See Hendry (1974) and Campos and Ericsson (1999) for empirical examples.

Table 1: Impulse indicator saturation and some extensions, as characterized by the variables involved.

Type	Name	Description	Variables	Definitions
1a	Impulse indicator saturation	zero-one dummies	$\{I_{it}\}$	$I_{it} = 1$ for $t = i$, zero otherwise
1b	Super saturation	step functions	$\{I_{it}, S_{it}\}$	$S_{it} = 1$ for $t \geq i$, zero otherwise
1c	Ultra saturation	broken linear trends	$\{I_{it}, S_{it}, T_{it}\}$	$T_{it} = t - i + 1$ for $t \geq i$, zero otherwise
1d	Sequential pairwise IIS	zero-one-one dummies	$\{P_{it}\}$	$P_{it} = 1$ for $t = i, i + 1$; zero otherwise
1e	Zero-sum pairwise IIS	“plus-one”– “minus-one” dummies	$\{Z_{it}\}$	$Z_{it} = +1$ for $t = i$, $Z_{it} = -1$ for $t = i + 1$, zero otherwise
2	Many many variables	more variables than observations	$\{x_{kt}\}$	$x_{kt} = \sum_{i=1}^T x_{kt} I_{it}$
3	Factors	factors, principal components	$\{f_{jt}\}$	$f_{jt} = \sum_{\forall k} w_{jk} x_{kt}$
4	Multiplicative indicator saturation	partial series	$\{x_{kt}^{(i)}, \forall i, k\}$	$x_{kt}^{(i)} = 0$ for $t < i$, $x_{kt}^{(i)} = x_{kt}$ for $t \geq i$

2. **Many many variables.** IIS provides a solution for dealing with more potential variables than observations, i.e., a set of K potential regressors $\{x_{kt}, k = 1, \dots, K\}$ for $K > T$. In the same spirit as IIS, block searches can be applied to a set of economic variables for which there are more variables than observations. Additionally, every economic series $\{x_{kt}\}$ is interpretable as the weighted sum of the impulse indicators $\{I_{it}\}$, where the weight on each impulse indicator I_{it} is the value of the economic series x_{kt} for the observation corresponding to the impulse indicator. Block searches across many many variables are thus interpretable as searches across particular, economically interesting combinations of impulse indicators. Empirical models may involve (say) K data aggregation assumptions with $K > T$ in practice; those assumptions can now be tested. Ericsson (2011a) proposes this test of data aggregation and applies it to aggregation assumptions in a global vector autoregression.
3. **Factors.** Factors and principal components are weighted sums of economic variables. So, from #2 above, factors and principal components are weighted sums of the impulse indicators. See Stock and Watson (2002, 2005), Bernanke, Boivin, and Elias (2005), and Castle, Clements, and Hendry (2011) for discussions.
4. **Multiplicative indicator saturation.** If a model's coefficient on x_{kt} is suspected to have changed at a particular date i , a natural way to capture that change is by including $S_{it}x_{kt}$ in the model, in addition to x_{kt} itself, with the coefficient on $S_{it}x_{kt}$ picking up the incremental change in the original coefficient on x_{kt} . If the break-point i is itself unknown, block searches with more potential variables than observations permit considering $S_{it}x_{kt}$ for *all* break-point dates i and variables k . This approach precisely nests the Andrews (1993) unknown breakpoint test and the Bai and Perron (1998) multiple breakpoint test, aside from directly allowing the error variance to alter. (IIS *does* allow the error variance to alter, but IIS is not very parsimonious in the way that it does so.)

To give a better sense of the dummy variables involved in Table 1, Figure 5 plots each of several types of dummies listed in Table 1 for $i = 2000$: specifically, I_{it} , S_{it} , T_{it} , P_{it} , Z_{it} , and $S_{it}x_{kt}$, where x_{kt} is the CBO forecast in Section 6.2. The saturation procedures listed in Table 1 search across all feasible dummies of a given type (or types); Figure 6 plots the sets of all feasible dummies for the specific dummy types plotted in Figure 5. Figures 5 and 6 thus emphasize how each saturation procedure focuses on parsimoniously representing certain sorts of alternatives.

While Table 1 details several interesting extensions of IIS, Table 1 is by no means an exhaustive list of extensions to IIS. Also, the choice of IIS-type procedure may itself be a combination of the entries in Table 1; and that choice may affect the power of the procedure to detect specific alternatives. For instance, the twenty-five impulse

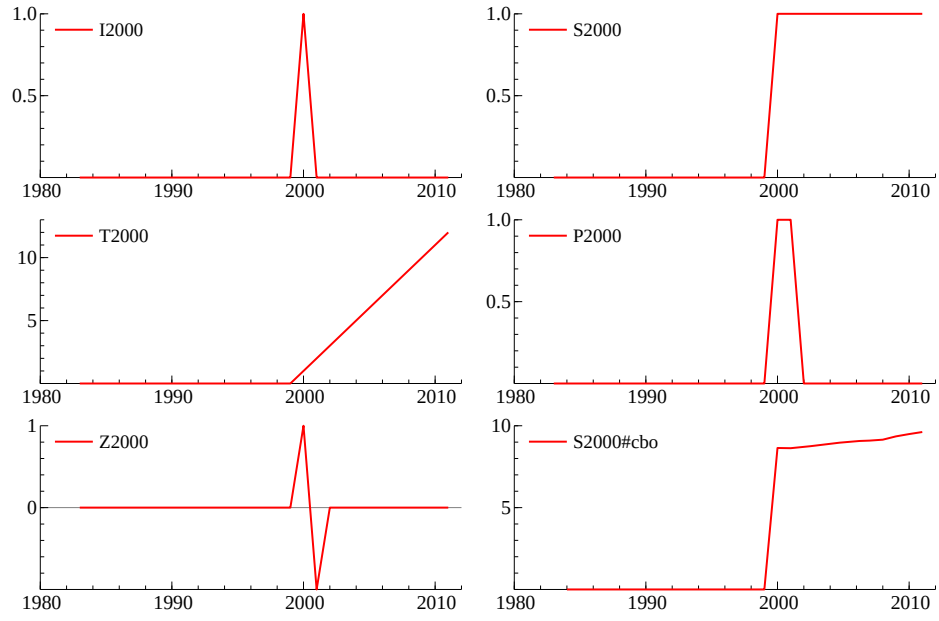


Figure 5: Examples of the dummies for IIS and its extensions.

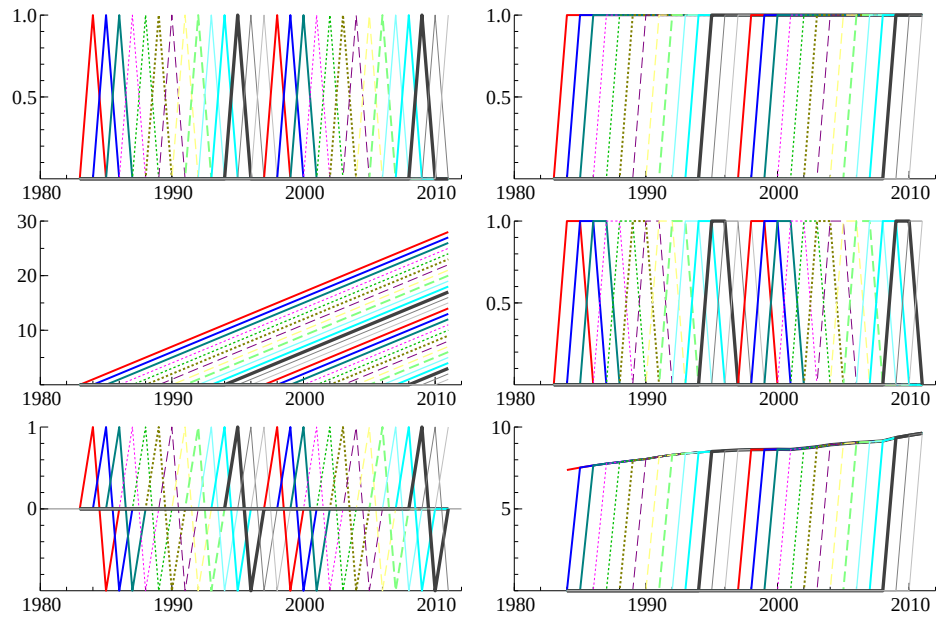


Figure 6: All dummies for IIS and certain classes of its extensions.

indicators $\{I_{it}, i = 76, \dots, 100\}$ are not a particularly parsimonious way of expressing a step shift that occurs three-quarters of the way through a sample of 100 observations (as in Example 2 above), whereas the single step dummy $S_{76,t}$ is.

4 Seasonally Unadjusted U.K. Money Demand

Ericsson, Hendry, and Tran (1994) (denoted EHT below) design an empirical model of U.K. narrow money demand, using seasonally unadjusted data. This section assesses the possible path dependence in selection of EHT's money demand model (Section 4.1), evaluates their model for parameter constancy using IIS (Section 4.2), and redesigns that model using Autometrics and IIS-type procedures (Section 4.3). Computer-automated model selection with Autometrics demonstrates the robustness of Ericsson, Hendry, and Tran's (1994) final equilibrium correction model to a wide range of feasible alternatives. Computer-automated model selection also improves on that model by using multi-path searches with IIS-type procedures.

4.1 Ericsson, Hendry, and Tran's (1994) Model

EHT analyze the measure of nominal narrow money M_1 (M), real total final expenditure (TFE) at 1985 prices (Y), the TFE deflator (P), and a net interest rate (R^{net}) that aims to capture the opportunity cost of holding money. Money and expenditure are in £ millions, the deflator is unity for 1985, and the interest rate is in fractions. The data are quarterly, 1963Q1–1989Q2. Allowing for lags and transformations, estimation is over 1964Q3–1989Q2, which is 100 observations ($T = 100$). For notational convenience, lowercase letters denote logs.

EHT test for and find a single cointegrating vector in a fifth-order vector autoregression of real money ($m-p$), real income (y), inflation (Δp), and the net interest rate (R^{net}), including a constant term and seasonal dummies (Q_{1t}, Q_{2t}, Q_{3t}). Furthermore, the variables y , Δp , and R^{net} appear to be weakly exogenous for that cointegrating vector, which is interpretable as a money demand relationship. EHT thus turn to modeling an equilibrium correction model of real money, starting with a fifth-order autoregressive distributed lag (ADL) in $m-p$, y , Δp , R^{net} , a constant term, and

seasonal dummies. EHT manually simplify that ADL to the following specification.

$$\begin{aligned}
\widehat{\Delta(m-p)}_t = & - \frac{0.95}{(0.12)} [\Delta_3(m-p)_{t-1}/3] - \frac{1.07}{(0.15)} [(\Delta p_t + \Delta p_{t-4})/2] \\
& + \frac{0.16}{(0.04)} \Delta^2 y_{t-2} \\
& - \frac{0.174}{(0.013)} (m-p-y)_{t-1} - \frac{1.189}{(0.093)} (\sum_{i=0}^2 R_{t-i}^{net}/3) \\
& + \frac{0.038}{(0.006)} - \frac{0.012}{(0.005)} Q_{1t} + \frac{0.010}{(0.005)} Q_{2t} + \frac{0.018}{(0.007)} Q_{3t} \quad (3)
\end{aligned}$$

$$\begin{aligned}
T = 100 [1964Q3-1989Q2] \quad R^2 = 0.84 \quad \hat{\sigma} = 1.348\% \quad Inn : F(24, 67) = 0.70 \\
AR : F(5, 86) = 1.03 \quad ARCH : F(4, 92) = 0.48 \quad Normality : \chi^2(2) = 5.63 \\
RESET : F(2, 89) = 0.81 \quad Hetero : F(13, 86) = 1.26 \quad Form : F(23, 76) = 0.98.
\end{aligned}$$

Equation (3) appears well-specified on the statistics reported, including tests of parameter constancy by recursive least squares. See Doornik and Hendry (2009) and Doornik (2009) for a detailed description of the diagnostic statistics in equation (3).

Autometrics—in Doornik and Hendry’s (2009) econometrics software package OxMetrics—implements automated model selection in a general-to-specific approach with diagnostic testing, multi-path search, and resolution of multiple terminal models through encompassing. Autometrics thus permits examining the possible path dependence of EHT’s model selection: EHT’s model in equation (3) arose from a manual general-to-specific model search that examined only a very limited number of potential simplification paths. Applying Autometrics at a 5% target size to a “naturally nesting” version of the unrestricted fifth-order ADL still obtains equation (3); see Ericsson (2010) on naturally nesting (and other types of) model representations. In obtaining equation (3), Autometrics considers 2^{24} (over 16 million) different potential models.¹ Hence, equation (3) is clearly robust to a vast range of feasible alternatives, and so is robust to the model selection path. Section 4.2 re-examines equation (3) with impulse indicator saturation, and Section 4.3 develops a new empirical model on the existing evidence, with both sections employing automated model selection.

Above, and in Section 4.2, automated model selection is implemented as a *diagnostic tool*, with the right-hand side variables from equation (3) forced (or “fixed”) to enter the estimated equations in the model search. If those variables are needed in the final selected model, then forcing them to enter the model helps guide the search procedure. If those variables are not needed, then that should be apparent in the final model from the insignificance of their coefficients. In Section 4.3, automated model

¹Although Autometrics considers 2^{24} distinct potential models, it actually estimates only 27 of those models in this instance, owing to the sophisticated nature of Autometrics’s search algorithm; see Doornik (2009) for details.

selection is implemented as a *model design tool*, albeit with the intercept, seasonal dummies, and long-run economic variables being fixed in the search process.

4.2 Model Evaluation With IIS

This subsection applies IIS-type procedures to equation (3) using Autometrics. These procedures detect anomalies in 1969Q1 and 1969Q2, where those anomalies are of the same sign and virtually equal magnitude. A sequential pairwise impulse indicator for those two quarters provides a concise summary. Testing is conducted at an approximate $1/K$ target size; see Doornik (2009) and Johansen and Nielsen (2009).

The results with IIS-type procedures can be summarized as follows. IIS at a 1% target level ($K = 100$) detects two impulse indicators—one for 1969Q1, and one for 1969Q2—each with a coefficient of about -0.04 , representing a growth rate of real money at about 4% per quarter lower than anticipated by the model. Super saturation ($K = 200$) does not detect any indicators at the 0.5% target level, but it does detect step dummies for 1969Q1 and 1969Q3 at the 1% target level, where the step dummies' coefficients are of opposite sign and virtually identical magnitude. Sequential pairwise IIS at the 0.33% target size ($K = 300$) detects a single paired impulse indicator (denoted $P_t^{1969q1q2}$) that is unity for 1969Q1 and 1969Q2, thus providing a convenient representation of the results from IIS and super saturation. Multiplicative IS at the 0.1% target size ($K = 1100$) likewise obtains that single paired impulse indicator.

No obvious changes in data measurement appear to have occurred in 1969Q1 and 1969Q2; see Clews, Healey, Hoggarth, and Mann (1990) and Topping and Bishop (1989). So, it remains to be determined what events induced a temporary decline in predicted real money in the first half of 1969, above and beyond the determinants already in equation (3).

4.3 Model Redesign with Autometrics and IIS

EHT (p. 207) note that equation (3) may be rewritten, re-interpreted, and (possibly) further simplified by transforming the dependent variable from the growth rate in real money (i.e., $\Delta(m-p)_t$) to the growth rate in nominal money (i.e., Δm_t). This section examines the implications of that transformation, using Autometrics to re-simplify the fifth-order ADL described in Section 4.1.

That ADL can be represented equivalently as an equation with the growth rate of nominal money as the dependent variable, even though the ADL determines the log-level of real money. Autometrics with multiplicative impulse saturation at a 0.1% target size ($K = 1025$) obtains the following simplification from that transformed ADL.

$$\begin{aligned}
\widehat{\Delta m_t} = & \left(\frac{\Delta_4 p_t}{4} - \frac{\Delta_3 m_{t-1}}{3} \right) + \frac{0.15}{(0.04)} \Delta^2(y+p)_{t-2} - \frac{0.042}{(0.009)} P_t^{1969q1q2} \\
& - \frac{0.161}{(0.008)} (m-p-y)_{t-1} - \frac{1.171}{(0.048)} (\sum_{i=0}^2 R_{t-i}^{net}/3) \\
& + \frac{0.044}{(0.005)} - \frac{0.011}{(0.004)} Q_{1t} + \frac{0.009}{(0.004)} Q_{2t} + \frac{0.015}{(0.006)} Q_{3t}
\end{aligned} \tag{4}$$

$$\begin{aligned}
T = 100 \text{ [1964Q3–1989Q2]} \quad R^2 = 0.82 \quad \hat{\sigma} = 1.23509\% \\
AR : F(5, 87) = 1.60 \quad ARCH : F(4, 92) = 1.89 \quad Normality : \chi^2(2) = 1.56 \\
RESET : F(2, 90) = 1.33 \quad Hetero : F(13, 86) = 2.09^* \quad Form : F(26, 73) = 1.46.
\end{aligned}$$

Equation (4) has a straightforward economic interpretation in line with standard *Ss*-type inventory models; see especially Miller and Orr (1966), Akerlof (1973, 1979), Akerlof and Milbourne (1980), Milbourne (1983), and Smith (1986). Nominal money growth equals the excess of annual inflation over nominal money growth in the past year, with additional adjustments from lagged money disequilibria, seasonality, and the acceleration of nominal income. Solved lag coefficients from equation (4) are very similar to those from equation (3). The dynamics of equation (4) is closely linked to the dynamic economic theory on which it is based. Specifically, the short-run dynamics of equation (4) is consistent with an *Ss*-type inventory model that has nominal short-run bounds, but where those bounds adjust to long-term effects.

Statistically, equation (4) is well-specified, apart from rejection by one test of heteroscedasticity. That rejection appears to arise from the inclusion of the sequential paired impulse indicator $P_t^{1969q1q2}$. The residual standard error in equation (4) is approximately 10% less than that in equation (3), primarily indicative of the statistical benefits of filtering the observations for 1969Q1–1969Q2.

Computer-automated model selection with Autometrics demonstrates the robustness of Ericsson, Hendry, and Tran’s (1994) final equilibrium correction model to a wide range of feasible alternatives. Computer-automated model selection also improves on that model by using multi-path searches with IIS-type procedures. Those searches would be tedious and prohibitively time-consuming with standard econometric packages.

5 Financial Markets in a GVAR

This section extends IIS results in Ericsson (2011a) and Ericsson and Reisman (2012a) for the global vector autoregressive model in Déés, di Mauro, Pesaran, and Smith (2007), focusing on equations for the U.S. equity market and the U.K. exchange rate.

Pesaran, Schuermann, and Weiner (2004) propose and implement global vector autoregressions (or GVARs) as an ingenious approach for capturing international linkages between country- or region-specific error correction models. Déés, di Mauro, Pesaran, and Smith (2007) extend that work to a larger number of countries and regions; and Pesaran, Schuermann, and Smith (2009) assess the forecasting properties of the GVAR implemented in DdPS.

The GVAR methodology has several attractive features:

- a versatile structure for characterizing international macroeconomic and financial linkages through multiple channels,
- a standardized economically appealing choice of variables (both domestic and foreign) for each country or region,
- a systematic treatment of long-run properties through cointegration analysis, and
- flexible dynamic specification through vector error correction modeling.

Ericsson (2011a) re-examines the theoretical and empirical underpinnings for global vector autoregressions (or GVARs), focusing on the following areas:

- treatment of exogeneity assumptions,
- parameter constancy, and
- data aggregation.

Ericsson (2011a) proposes extensions and improvements in these areas, with the aim of achieving an even more robust approach to GVAR modeling.

With that in mind, Ericsson (2011a) and Ericsson and Reisman (2012a) demonstrate these refinements to GVAR modeling by calculating IIS for the GVAR in Déés, di Mauro, Pesaran, and Smith (2007), focusing on the VARX* specifications for the United States, the euro area, the United Kingdom, Sweden, Brazil, Japan, and China. The GVAR in Déés, di Mauro, Pesaran, and Smith (2007) is estimated over 1979Q4–2003Q4 ($T = 97$) and, even with a target level of 0.1%, IIS detects numerous breaks, indicating scope for improving this GVAR. The dates of the retained impulses suggest specific interpretations regarding crises, jumps, and changes in regime in these economies; and the presence of these impulse dummies indicates directions to pursue for improving this GVAR. The remainder of this section examines the equations for the U.S. equity market and the U.K. exchange rate from the GVAR in Déés, di Mauro, Pesaran, and Smith (2007), comparing and interpreting several of the IIS-type procedures listed in Table 1.

For equity prices in the United States, IIS at a 0.1% target level detects an impulse dummy for 1987Q4, reflecting the sharp decline in the stock market on Black Monday (October 19, 1987). Expanding on that result, Table 2 reports the dummies detected by other IIS-type procedures in Table 1, with target sizes adjusted according to the number of potential regressors being considered.

Table 2: Coefficients and estimated standard errors for indicators detected in the GVAR equation for U.S. stock prices.

Type	IIS	Super	Ultra	Pairwise	Zero-sum	Multi-	Reference
K	97	194	291	388	485	1940	3
α	0.1%	0.05%	0.033%	0.025%	0.02%	0.01%	—
I1987Q4	−0.275 (0.059)						−0.272 (0.059)
I1988Q1							0.064 (0.072)
I1988Q2							−0.028 (0.065)
T1987Q4			−0.276 (0.059)	✓	✓		
T1988Q1			0.568 (0.118)	✓	✓		
T1988Q2			−0.292 (0.060)	✓	✓		
$\hat{\sigma}$ (%)	5.453	6.107	5.452	5.452	5.452	6.107	5.487

Numerically, the three broken trends for 1987Q4, 1988Q1, and 1988Q2 virtually replicate the inclusion of the single impulse dummy for 1987Q4. For reference, the final column reports the GVAR equation for U.S. equity prices, with impulse dummies included for 1987Q4, 1988Q1, and 1988Q2. The impulse dummies in the reference equation virtually “unwind” the effects of the broken trends and show that the dominant effect is from a single downward shift in equity prices in 1987Q4. These results appear qualitatively different but are numerically similar, highlighting some challenges to automated model selection when there are many many potential explanatory variables.

For the GVAR’s equation for the UK exchange rate, IIS retains an impulse indicator dummy for 1992Q4, reflecting the substantial devaluation of the pound sterling around the time of Black Wednesday (September 16, 1992), when the British government was forced to withdraw the pound sterling from the ERM. Expanding on that result, Table 3 reports the dummies detected by other IIS-type procedures, with target sizes adjusted according to the number of potential regressors being considered. Numerically, the two step functions for 1992Q4 and 1993Q1 virtually replicate the inclusion of the single impulse dummy for 1992Q3. However, as the “reference” coefficients in the last column indicate, 1992Q3 is accompanied by some run-up in the pound sterling.

Table 3: Coefficients and estimated standard errors for indicators detected in the GVAR equation for U.K. exchange rates.

Type	IIS	Super	Ultra	Pair	Zero-sum	Multi-	Reference
K	97	194	291	388	485	1940	3
α	0.1%	0.05%	0.033%	0.025%	0.02%	0.01%	—
I1992Q3							−0.084 (0.031)
I1992Q4	0.157 (0.032)						0.155 (0.031)
I1993Q1							−0.012 (0.034)
S1992Q4		0.159 (0.033)	✓	✓		✓	
S1993Q1		−0.155 (0.033)	✓	✓		✓	
Z1992Q3					−0.119 (0.022)		
$\hat{\sigma}$ (%)	2.968	2.980	2.980	2.980	2.882	2.980	2.871

6 Two Additional Applications

This section considers two additional recent applications of IIS and IIS-type procedures. Section 6.1 summarizes results from Ericsson and Chekmasova (2012) in detecting regime changes in the U.S. Taylor rule, and Section 6.2 describes the reinterpretation in Ericsson (2012) of IIS-type tests as tests of time-varying bias.

6.1 Regime Changes and the U.S. Taylor Rule

Taylor (1993) proposes an interpretation of Fed monetary policy as weighing inflation and the output gap when setting the federal funds rate; see also McCallum (1993). Judd and Rudebusch (1998) and others have noted that, empirically, the Taylor rule appears to be nonconstant over time. Ericsson and Chekmasova (2012) apply IIS and multiplicative indicator saturation to an empirical Taylor rule and find that the coefficients in a Taylor rule change markedly with Federal Reserve Board chairmen. Even within a given chairmanship, temporary shifts in coefficients can be sizeable.

6.2 Tests of Forecast Bias

In light of Sections 2 and 3, Ericsson (2012) reinterprets the Mincer–Zarnowitz test of forecast bias as a very special case of super saturation in which only S_{1t} is retained.

Conversely, that Mincer–Zarnowitz test is interpretable as an IIS test, but where equality across the impulse coefficients is *imposed* (or assumed), and the hypothesis is that *an* impulse coefficient is zero. The IIS-based tests of forecast bias also generalize that Mincer–Zarnowitz test by allowing for time-varying bias. These observations highlight the strength of the Mincer–Zarnowitz test (that it focuses on detecting a constant nonzero forecast bias) and also its weakness (that it assumes that the forecast bias is constant over time).

Ericsson (2012) finds that tests based on impulse indicator saturation detect economically large, persistent, and highly statistically significant time-varying biases in Federal agencies’ forecasts of the U.S. gross Federal debt, particularly for 1990, 2001–2003, and 2008–2011. Biases differ across different agencies’ forecasts as well as across time. The IIS approach also explains *why* standard Mincer–Zarnowitz tests fail to detect bias in these forecasts of U.S. gross Federal debt.

7 Remarks

Computer-automated model selection and impulse indicator saturation are highly valuable for the detection and quantification of crises, jumps, and changes in regime. This paper proposes extensions to impulse indicator saturation that may provide more power to detect breaks in common economic situations. These tools are applicable to analyzing a wide class of models, and even of (e.g.) forecasts that have no formal model.

Impulse indicator saturation defines a generic procedure for detecting and quantifying crises, jumps, and changes in regime. Such structural breaks can be persistent yet time-varying; they can be difficult to detect in a timely fashion; and they may have substantive implications for policy analysis. The IIS approach aims to address these issues, with extensions of IIS aiming to characterize systematic properties across retained impulses. The IIS approach also links directly to techniques for robustifying forecasts, noting that intercept correction is a variant of super saturation; see Clements and Hendry (1996, 1999, 2002), Hendry (2006), and Castle, Fawcett, and Hendry (2010).

IIS-based tests may detect not only “location shifts”, but also outliers and other forms of mis-specification, such as changes in the forecast error variance. For outliers and changes in error variance in particular, retained impulses would not typically be sequential or (for adjacent impulses) of the same sign and similar magnitude. These caveats emphasize the usefulness of super saturation and other extended forms of IIS in characterizing the persistence of structural breaks.

IIS-based tests can serve as diagnostic tools to detect what’s wrong with a model, and as developmental tools to suggest how that model may be improved. While rejection of the null doesn’t imply the alternative, the *date*-specific nature of IIS-type

procedures when applied to time series data can aid in identifying the sources of a model’s weaknesses, and hence ways in which a model can be improved. Use of these tests in model development is consistent with a progressive modeling approach; see White (1990).

The IIS approach has many potential applications, beyond its initial roles in model evaluation and robust estimation. As discussed above, it can be used for detecting crises, jumps, and changes in regime. IIS also provides a framework for creating near-realtime early-warning and rapid-detection devices, e.g., of financial market anomalies. Finally, IIS generalizes to systems.

Hendry (2011) sketches out how recent econometric tools and software developments have made empirical modeling an exciting and vibrant enterprise in economics. Additional results in the statistical theory of model selection—and corresponding software implementation—will no doubt further aid this endeavor. The current paper serves to exemplify, extend, and illustrate some aspects of how such empirical modeling might proceed.

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